

WILFRID LAURIER UNIVERSITY

DEPARTMENT OF GEOGRAPHY AND ENVIRONMENTAL STUDIES

GG468 – Advanced Spatial Analysis

LAB ASSIGNMENT 6

Instructor: Dr. Erin Li

Scope:

This assignment introduces students to geostatistical interpolation using Ordinary Kriging in ArcGIS Pro. Students will explore spatial point data, evaluate spatial autocorrelation through semivariogram analysis, fit an appropriate theoretical semivariogram model, generate an Ordinary Kriging prediction surface, and compare the resulting interpolation to Inverse Distance Weighting (IDW). The lab emphasizes both conceptual understanding and practical GIS implementation for continuous environmental variables such as rainfall.

Steps:

Part I: Project Setup and Data Exploration

1. Create a new project in ArcGIS Pro and name it:
yourname_A6.aprx
2. Load the provided rainfall station dataset (Rainfall_Stations.shp) from:
Module 6 > Lab Assignment > Data_Assignment_6
3. Select the basemap **Topographic** or **Light Gray Canvas**.
4. Examine the attribute table of the rainfall stations dataset.
 - Identify station locations
 - Review rainfall measurements (mm)
 - Inspect for missing values or obvious data entry errors
5. Visualize rainfall station values using graduated symbols.
6. Perform exploratory spatial data analysis:
 - Histogram of rainfall values
 - Summary statistics
 - Spatial distribution of points
 - Identify potential outliers
 - Evaluate whether rainfall appears continuous and spatially autocorrelated

Part II: IDW Baseline Surface

7. Use the Geoprocessing Tool: **Spatial Analyst Tools > Interpolation > IDW**
8. Generate an IDW rainfall surface using default parameters.
9. Export the IDW output as a raster layer for comparison.

Part III: Ordinary Kriging Workflow

10. Open the **Geostatistical Wizard** from the **Analysis** tab. If the tool is unavailable, check that the required geostatistical analysis license is active.
11. Before running Ordinary Kriging, examine whether a broad spatial trend exists in the rainfall data. In the Geostatistical Wizard, select **Global Polynomial Interpolation** under **Deterministic Methods**, set the rainfall station layer as the input dataset, and choose the rainfall field as the data field.
12. Preview the trend surface using a first-order polynomial. If appropriate, increase the polynomial order to 2 to check whether a broad curved trend is present. Use this step only to identify large-scale trend, not to model local variation.
13. Review the cross-validation plot and RMSE for the trend model. Record whether a broad trend appears to exist in the rainfall data.
14. Return to the first Geostatistical Wizard window. Select **Kriging/CoKriging** and set the rainfall station layer and rainfall field as the source dataset and data field.
15. Choose **Ordinary Kriging** and set the output type to **Prediction**.
16. If a broad trend was identified earlier, set the **Trend Removal Order** to the appropriate polynomial order, such as **Second Order**. If no clear broad trend is present, keep trend removal at **None**.
17. Continue to the semivariogram window. Examine the empirical semivariogram and fitted model, including:
 - Nugget
 - Range
 - Sill
 - Binned semivariance values
18. Select an appropriate semivariogram model, such as **Spherical**, **Exponential**, or **Gaussian**. Use the **Optimize** option after selecting the model to estimate suitable model parameters.
19. Review the prediction preview and search neighborhood settings. Keep the default search neighborhood settings unless there is a clear reason to modify them.
20. Review the final cross-validation results and record:
 - Mean Error
 - Root Mean Square Error
 - Average Standard Error
 - Standardized RMSE
21. Click **Finish** to create the Ordinary Kriging geostatistical layer.
22. Export the geostatistical layer to a permanent raster:
Contents pane > Right-click Kriging layer > Export Layer > To Rasters
23. In the **GA Layer to Rasters** tool, save the output raster as:
ordinary_kriging_prediction.tif
24. Set an appropriate output cell size and use the study area boundary as the processing extent and mask, if available. Then click **Run**.
25. Symbolize the final raster using a stretched color ramp and save the project.

Part IV: Visualization and Interpretation

18. Create Map Layout [1] showing:

- Rainfall stations
- IDW interpolation surface
- Legend, scale bar, north arrow, title

19. Create Map Layout [2] showing:

- Rainfall stations
- Ordinary Kriging prediction surface
- Semivariogram model summary (model type + nugget, range, sill)
- Legend, scale bar, north arrow, title

20. Create Map Layout [3] showing:

- Kriging standard error (uncertainty) map
- Areas of high vs. low prediction confidence
- Legend, scale bar, north arrow, title

Part V: Comparison and Reflection

21. Compare IDW and Ordinary Kriging outputs by addressing:

- How do prediction surfaces differ?
- Where does Kriging produce smoother or more realistic estimates?
- Where is uncertainty highest, and why?
- Why might distance alone be insufficient?

Deliverables:

a. PDF Report containing:

- Map Layout [1]: IDW Surface
- Map Layout [2]: Ordinary Kriging Surface
- Map Layout [3]: Kriging Standard Error Surface
- Semivariogram graph screenshot
- Cross-validation statistics table
- Short comparison discussion (0.5 page)

b. ArcGIS Pro Project Package:

- .aprx file
- Geodatabase
- All generated rasters/layers

Submission:

Upload all documents to:

D2L > Assessments > Dropbox > Lab_Assignment_6

Submission due: Nov 12 at 12:00 pm ET

Evaluation Rubric (100 Marks)

Component	Criteria	Marks
Data Exploration	Correct examination of rainfall data, summary statistics, histogram, and identification of data characteristics	10
IDW Interpolation	Successful generation of IDW surface and appropriate visualization	10
Ordinary Kriging Workflow	Correct execution of Geostatistical Wizard workflow, trend assessment, model fitting, and raster generation	25
Semivariogram Analysis	Appropriate interpretation of empirical semivariogram, selection of theoretical model, and reporting of nugget, range, and sill	15
Cross-Validation Assessment	Accurate reporting and interpretation of validation statistics (ME, RMSE, ASE, Standardized RMSE)	10
Map Layout Quality	Professional cartographic design including legends, scale bars, north arrows, titles, and clear symbology	15
Comparison & Reflection	Critical comparison of IDW and Ordinary Kriging, discussion of uncertainty, and reflection on interpolation method selection	10
Report Organization & Presentation	Clarity, completeness, formatting, grammar, and overall professionalism	5
Total		100

Appendix: Geostatistical Wizard – Kriging/CoKriging

